

~~Functions~~

Lecture 2

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A quick recap

- Conditional: $p \rightarrow q$.
- Contrapositive: $\neg q \rightarrow \neg p$
- Subset: $A \subset B$ is essentially a conditional of $x \in A \rightarrow x \in B$.
- Extensionality: $A = B$ means $A \subset B$ and $B \subset A$.

Recap examples

Lec 1 HW Q1

Recap examples

$\mathbb{Z} = 2\mathbb{Z}$? Prove or disprove.

Recap examples

If $\mathbb{Z} = n\mathbb{Z}$ for some integer n , then n is either 1 or -1 . Prove or disprove.

cf. Lec 1 HW Q7

Recap examples

$\{xy \mid x, y \in \mathbb{Z}\} = \mathbb{Z}$? Prove or disprove.

cf. PCSA Q2

Recap examples

$49\mathbb{Z} + 10\mathbb{Z} = \mathbb{Z}$? Prove or disprove.

cf. Lec 1 HW Q4

Recap examples

If $x^n \neq 1$, then $x^m \neq 1$ for any factors m of n . Prove or disprove.

cf. Lec 1 HW Q5

Function

Well-defined function

Definition 1 (Function)

A *function* consists of three data:

1. A set, called the *domain*.
2. A set, called the *codomain*.
3. An assignment with the following property:
 1. Every element in the domain is assigned an element in the codomain.
 2. Such an assignment is unique.



Definition 2 (Terminology & Notation)

$f : X \rightarrow Y$; f is a function whose domain is X and codomain is Y .

Image: The elements in the codomain that are assigned by f .

Preimage: The elements in the domain that are assigned by f .

$$f(x) = y$$

$$x \mapsto y$$



Examples for well?defined functions

1. $f : \mathbb{Q} \rightarrow (\mathbb{Z} \times \mathbb{Z})$, If x can be written as $\frac{p}{q}$, then $f(x) = (p, q)$.
2. $f : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$, If x can be written as (x_1, x_2) , then $f(x) = x_1$.
3. $f : \mathbb{Z}^+ \rightarrow \mathbb{C}$, For $n \in \mathbb{Z}^+$, if $z \in \mathbb{C}$ satisfies $z^n = 1$, then $f(n) = z$.
4. $f : \mathbb{C} \rightarrow \mathbb{C}/=$, If X contains only one element x , then $f(x) = X$.
5. $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = \tan(x)$.
6. $f : \mathbb{Z} \rightarrow \mathbb{Z}$, For $x \in \mathbb{Z}$, if y is a multiple of x , then $f(x) = y$.
7. $f : \mathbb{Z} \rightarrow \{\emptyset\}$, $f(x) = \emptyset$.
8. $f : \mathbb{Z} \rightarrow \{n\mathbb{Z} \mid n \in \mathbb{Z}\}$, $f(x) = x\mathbb{Z}$.

Equality of functions

Definition 1 (Equality of functions)

Two functions $f, g : X \rightarrow Y$ are considered to be *equal* if for any x in the domain, $f(x) = g(x)$ holds.



Identity function

Definition 1 (Identity function)

An *identity function* of a set X is a function $f : X \rightarrow X$ such that $f(x) = x$ for all x in X .



Notations

- $1_X : X \rightarrow X$
- $I_X : X \rightarrow X$
- $\text{id}_X : X \rightarrow X$

Bonus: How many functions exist between two sets?

cf. PCSA Q16

Let X and Y be two finite sets, with the size n and m resp.

Injection

Definition 1 (Injection)

A function f is said to be *injective* or called an *injection* if for any elements $x \neq y$ in the domain, $f(x) \neq f(y)$ holds.



Definition 2 (Injection - equivalent)

A function f is said to be *injective* or called an *injection* if for any elements x and y , $f(x) = f(y)$ then $x = y$.



Injection

Examples

1. $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2$
2. $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = \sin(x)$
3. $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = a + x$ for any $a \in \mathbb{R}$
4. $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = ax$ for any $a \in \mathbb{R}$
5. $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = e^x$
6. $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = |x|$
7. $f : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}, f(x, y) = x + y$

Surjection

Definition 1 (Surjection)

A function f is said to be *surjective* or called a *surjection* if for any element y in the codomain, there exists an element x in the domain such that $f(x) = y$.



Surjection

Examples

1. $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2$
2. $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = \sin(x)$
3. $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = a + x$ for any $a \in \mathbb{R}$
4. $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = ax$ for any $a \in \mathbb{R}$
5. $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = e^x$
6. $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = |x|$
7. $f : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}, f(x, y) = x + y$

Bonus: Coincidence of two in finite cases

Theorem 1

For two finite sets X and Y such that $|X| = |Y|$, a function $f : X \rightarrow Y$ is injective if and only if surjective.



cf. PCSA Q20.

Bijection

Definition 1 (Bijection)

A function f is said to be *bijection* or called a *bijection* if it is both injective and surjective.



Relationship between functions and sizes of sets

For two sets X and Y ,

- If there exists an injection $f : X \rightarrow Y$, then $|X| \leq |Y|$.
- If there exists a surjection $f : X \rightarrow Y$, then $|X| \geq |Y|$.
- If there exists a bijection $f : X \rightarrow Y$, then $|X| = |Y|$.

Bonus: How many bijections exist?

Let X and Y be two finite sets, with the size n .

Example:

- A bijection on three-element sets

Composition

Definition 1 (Composition)

For two functions $f : X \rightarrow Y$ and $g : Y \rightarrow Z$, the *composition of f and g* is a function $g \circ f : X \rightarrow Z$ given by $(g \circ f)(x) = g(f(x))$.



Notes

- Be careful with the order.
- Does the composition always exist between any two functions?
- Do they commute?

Examples

1. $\alpha : \mathbb{R} \rightarrow \mathbb{R}, \beta : \mathbb{R} \rightarrow \mathbb{R}, \alpha(x) = x^2 + 2, \beta(x) = x - 1$
2. $1_X : X \rightarrow X, 1_X : X \rightarrow X, 1_{X(x)} = x, 1_{X(x)} = x$

Function

Q33

Inverses

Definition 1 (Inverse of a function)

The *inverse* of a function $f : X \rightarrow Y$ is a function $f^{-1} : Y \rightarrow X$ such that $f^{-1} \circ f = 1_X$ and $f \circ f^{-1} = 1_Y$.



Examples

- $f : \mathbb{R} \rightarrow \mathbb{R}, g : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x + 5, g(x) = x - 5$
- $f : \mathbb{N} \rightarrow \mathbb{N}, g : \mathbb{N} \rightarrow \mathbb{N}, f(x) = x + 5, g(x) = |x - 5|$

Invertible iff bijective

Theorem 1

A function is invertible if and only if it is bijective.



Binary operations[이항 연산]

Definition

Definition 1 (Binary operation)

A *binary operation* of a set X is an assignment such that:

1. For every two elements in the set, an element of the set is assigned.
2. Such an assignment is unique.



Notation: $(X, \circledast)$, $x \circledast y$

Examples of binary operations

Well-defined or not?

1. Positive integers, addition. $(\mathbb{Z}^+, +)$
2. Integers, addition. $(\mathbb{Z}, +)$
3. Real numbers, addition. $(\mathbb{R}, +)$
4. Integers, subtraction. $(\mathbb{Z}, -)$
5. Positive integers, subtraction. $(\mathbb{Z}^+, -)$
6. Integers, multiplication. $(\mathbb{Z}, \times)$
7. Real numbers, division. $(\mathbb{R}, /)$
8. Real numbers, exponentiation. $(\mathbb{R}, ^)$

Examples of binary operations

1. Real numbers, larger one between two numbers. $(\mathbb{R}, \max)$
2. Integers, greatest common divisor. $(\mathbb{Z}, \gcd)$
3. Integers, remainder. $(\mathbb{Z}, \text{mod})$
4. Strings, concatenation. $(\Sigma^*, +)$
5. Subsets of a set X , intersection. $(\mathcal{P}(X), \cap)$
6. Subsets of a set X , Cartesian product. $(\mathcal{P}(X), \times)$
7. Matrices, multiplication. $(\text{GL}_{\mathbb{R}(n)}, \times)$
8. Three-dimensional Euclidean vector space, inner product. $(\mathbb{R}^3, \cdot)$
9. n -th roots of unity, multiplication. $(\mu_n, \times)$

Identity element[항등원]

$$x \circledast e = x \text{ and } e \circledast x = x$$

Notice & Solution time